BRIEF REPORT

Validation of the Sybil Lung Cancer Risk Prediction Model in a Predominantly Black Screening Cohort: A Brief Report

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ABSTRACT

Introduction: Lung cancer remains the leading cause of cancer-related mortality in the United States and worldwide. Black Americans have a higher lung cancer incidence and mortality risk despite lower cumulative tobacco exposure. Sybil is an open-source deep learning model that predicts future lung cancer risk from a single low-dose computed tomography (LDCT). Sybil was trained using National Lung Screening Trial data, in which more than 90% of the participants were White, and Sybil's performance in diverse populations remains unclear.

Methods: We conducted a retrospective cohort study of 5883 LDCTs from 3177 participants (2014–2025). Sybil generated risk predictions for 1 to 6 years. Model performance was assessed on the basis of the area under the receiver operating characteristic curve.

Results: This cohort included 64.9% Black individuals. Among 105 lung cancers, 61% were diagnosed within 1 year. Sybil reported strong discrimination for baseline LDCTs, with areas under the receiver operating

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characteristic curve ranging from 0.91 at 1 year (95% confidence interval: 87.7–94.6) to 0.82 at 6 years (95% confidence interval: 77.6–86.1), with comparable performance between Black and non-Black individuals.

Conclusions: Sybil reported robust lung cancer risk prediction in this screening cohort, supporting its generalizability and highlighting its potential to improve risk-based screening beyond the current eligibility criteria in diverse populations.

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Keywords: Lung cancer screening; Risk prediction; Sybil; Artificial intelligence; Early detection; Precision screening

Introduction

Lung cancer is characterized by persistent race disparities in outcomes.[1] In the United States, non-Hispanic Black individuals experience higher lung cancer incidence and mortality than White individuals despite lower cumulative tobacco exposure.[2] Identification as Black or African American may reflect exposure to the cumulative effects of residential segregation, socioeconomic inequities, disproportionate environmental and occupational exposures, and unequal access to preventive and specialty health care.

The National Lung Screening Trial enrolled individuals aged 55 to 74 years with a smoking history of 30 pack-years or more who currently smoked or had quit within 15 years and reported a 20% reduction in lung cancer mortality with low-dose computed tomography (LDCT) screening and a lung cancer detection rate of 1.1%.[3] These findings formed the basis for subsequent United States Preventive Services Task Force (USPSTF) screening recommendations. In 2021, the USPSTF expanded eligibility by lowering the starting age from 55 to 50 years and the minimum tobacco exposure from 30 to 20 pack-years to increase screening eligibility and reduce racial and ethnic disparities associated with the original criteria.[4]

Nevertheless, current USPSTF criteria still fail to identify many high-risk individuals, particularly among racial and ethnic populations that are historically underrepresented in lung cancer screening (LCS) trials.[5–7] This has prompted growing interest in individualized risk-prediction approaches to improve both the efficiency and equity of LCS.

Sybil is an open-source deep learning model that predicts future lung cancer risk from single LDCT axial images without requiring clinical or demographic inputs and has reported accurate risk stratification in prior cohorts.[8–10] Sybil's output consists of six numerical risk scores (range: 0.0–1.0), representing the cumulative future lung cancer risk at each year (1–6 y) after the analyzed LDCT scan.

Sybil was trained using National Lung Screening Trial data, in which all participants were US-based and more than 90% were White.[3] It was subsequently validated in predominantly White and Asian cohorts.[8,10] Its performance in racially diverse US screening populations has not been tested. In this study, we evaluated Sybil's performance in a predominantly Black LCS cohort within an urban safety-net health system that primarily serves individuals with low incomes, limited insurance coverage, and barriers to healthcare access. We aimed to assess model discrimination across multiple time horizons and examine the consistency of its performance across racial subgroups and using longitudinal LDCT imaging.

Materials and Methods

We conducted a retrospective cohort study of 5883 LDCTs from 3177 participants enrolled in the University of Illinois Chicago (UI Health) LCS program between 2014 and 2025, including 105 individuals subsequently diagnosed with lung cancer. The LCS was performed as the standard of care and followed the USPSTF eligibility criteria, with some individuals screened outside these thresholds on the basis of clinician judgment. Demographic characteristics and smoking history were collected at program entry and verified at each LDCT encounter. Lung cancer diagnoses were pathologically confirmed and recorded.

The Sybil open-source model (version 1.4.0) was downloaded from the public GitHub (https://github.com/reginabarzilaygroup/Sybil) repository and was not retrained or modified. Sybil was applied to each LDCT, providing lung cancer risk estimates for 1 to 6 years. Model discrimination was assessed using the area under the receiver operating characteristic curve (AUC). Analyses included baseline and follow-up LDCTs stratified by race. The 95% confidence intervals (CIs) for the AUCs were calculated using DeLong's nonparametric method.[11] This study was approved by the University of Illinois Chicago Institutional Review Board (Institutional Review Board protocol No. 2026–0143).

Results

The cohort was 54.1% male and 45.9% female; 64.9% identified as Black, 19.2% as White, 4% as Asian, 11.9% as other or unknown, and 13% as Hispanic or Latino (Table 1). Insurance coverage reflected a safety-net population: 40% with Medicaid (US public insurance for individuals with low incomes), 40% with Medicare

Table 1. Characteristics of the Study Cohort

Patient Groups	Overall Cohort, Total No. (%)	Black Cohort, Total No. (%)	Overall Cohort; Future Cancers Diagnosed, No. (%)	Black Cohort; Future Cancers Diagnosed, No. (%)
No. of participants	3177 (100.0)	2061 (100.0)	105 (100.0)	72 (100.0)
Age, y[a]				
<40	4 (0.1)	2 (0.1)	0 (0.0)	0 (0.0)
40-50	30 (0.9)	19 (0.9)	0 (0.0)	0 (0.0)
50-60	996 (31.4)	629 (30.5)	19 (18.1)	14 (19.4)
60-70	1687 (53.1)	1127 (54.7)	67 (63.8)	46 (63.9)
70-80	446 (14.0)	278 (13.5)	19 (18.1)	12 (16.7)
>80	14 (0.4)	6 (0.3)	0 (0.0)	0 (0.0)
Sex				
Female	1457 (45.9)	1104 (53.6)	55 (52.4)	41 (56.9)
Male	1720 (54.1)	957 (46.4)	50 (47.6)	31 (43.1)
Race[b]				
White	611 (19.2)	0 (0.0)	24 (22.9)	0 (0.0)
Black or African American	2061 (64.9)	2061 (100.0)	72 (68.6)	72 (100.0)
Asian	126 (4.0)	0 (0.0)	3 (2.9)	0 (0.0)
Other/Unknown	379 (11.9)	0 (0.0)	6 (5.7)	0 (0.0)
Ethnicity[c]				
Hispanic or Latino	413 (13.0)	8 (0.4)	8 (7.6)	1 (1.4)
Not Hispanic or Latino	2699 (85.0)	2024 (98.2)	96 (91.4)	71 (98.6)
Other/Unknown	65 (2.0)	29 (1.4)	1 (1.0)	0 (0.0)
Insurance				
Private	597 (18.8)	362 (17.6)	18 (17.1)	13 (18.1)
Medicare	1021 (32.1)	661 (32.1)	45 (42.9)	30 (41.7)
Medicaid	1255 (39.5)	832 (40.4)	35 (33.3)	24 (33.3)
Medicare andMedicaid	284 (8.9)	193 (9.4)	5 (4.8)	4 (5.6)
Selfpay/uninsured	11 (0.3)	5 (0.2)	2 (1.9)	1 (1.4)
Other	9 (0.3)	8 (0.4)	0 (0.0)	0 (0.0)
Cancer stage at diagnosis				
I			51 (48.6)	36 (50.0)
II			16 (15.2)	10 (13.9)
III			24 (22.9)	18 (25.0)
IV			14 (13.3)	8 (11.1)
Smoking status[a,d]				
Never	35 (1.1)	16 (0.8)	0 (0.0)	0 (0.0)
Former	1147 (36.1)	682 (33.1)	36 (34.3)	17 (23.6)
Current	1995 (62.8)	1363 (66.1)	69 (65.7)	55 (76.4)
Pack-y smoking history range[a,d]				
<20	382 (12.0)	254 (12.3)	3 (2.9)	3 (4.2)
20-30	948 (29.8)	680 (33.0)	27 (25.7)	22 (30.6)
30-40	715 (22.5)	445 (21.6)	20 (19.0)	11 (15.3)
40-50	603 (19.0)	378 (18.3)	30 (28.6)	22 (30.6)
50-60	267 (8.4)	171 (8.3)	14 (13.3)	11 (15.3)
60-70	110 (3.5)	63 (3.1)	4 (3.8)	1 (1.4)
70-80	43 (1.4)	22 (1.1)	3 (2.9)	0 (0.0)
80-90	47 (1.5)	15 (0.7)	2 (1.9)	1 (1.4)
90-100	19 (0.6)	11 (0.5)	1 (1.0)	1 (1.4)
>100	43 (1.4)	22 (1.1)	1 (1.0)	0 (0.0)
USPSTF eligibility[a,e]				
eligible	2680 (84.4)	1743 (84.6)	102 (97.1)	69 (95.8)
ineligible	497 (15.6)	318 (15.4)	3 (2.9)	3 (4.2)

(continued)

Table 1. Continued

Patient Groups	Overall Cohort, Total No. (%)	Black Cohort, Total No. (%)	Overall Cohort; Future Cancers Diagnosed, No. (%)	Black Cohort; Future Cancers Diagnosed, No. (%)
Time to cancer diagnosis or last negative follow-up, y				
1	710 (22.3)	452 (21.9)	64 (61.0)	42 (58.3)
2	757 (23.8)	488 (23.7)	12 (11.4)	11 (15.3)
3	513 (16.1)	317 (15.4)	8 (7.6)	5 (6.9)
4	360 (11.3)	227 (11.0)	5 (4.8)	4 (5.6)
5	178 (5.6)	120 (5.8)	7 (6.7)	4 (5.6)
6	146 (4.6)	106 (5.1)	2 (1.9)	1 (1.4)
>6	513 (16.1)	351 (17.0)	7 (6.7)	5 (6.9)

CT, computed tomography; USPTF, United States Preventive Services Task Force.
[a]Screening eligibility primarily followed USPSTF criteria (2013: aged 55-80 y, ≥30 pack-y, current use or quit within 15 y; 2021: aged 50-80 y, ≥20 pack-y, current use or quit within 15 y). Nevertheless, some individuals outside these thresholds (i.e., age <50 y, <20 pack-y, or who never smoked) were screened on the basis of clinician judgment.
[b]Race was self-reported in the electronic medical record. Asian subgroups were categorized as Asian, and participants who were identified as "Black or African American" were categorized as Black.
[c]Ethnicity was self-reported in the electronic medical record and categorized as Hispanic or Latino or not Hispanic or Latino.
[d]Smoking history (never, current, or former) and pack-years were verified at each low-dose CT examination; discrepancies with values documented in the electronic medical record were audited, and participant-reported values were used for this analysis.
[e]USPSTF eligibility was determined using the criteria in effect at the time of screening. Participants who did not meet multiple eligibility criteria were counted only once in the ineligible category. The eligible and ineligible groups were mutually exclusive and sum to the total participant count.

(US public insurance, primarily for adults aged ≥65 y), 19% with private insurance (employer-sponsored or individually purchased insurance), and 1% with self-pay, uninsured status, or other. Among the 105 participants who were subsequently diagnosed with lung cancer, 61.0% were diagnosed within 12 months of baseline LDCT, 11.4% within 1 to 2 years, 12.4% within 2 to 4 years, and 15.3% after more than 4 years. Approximately 50% of lung cancers were diagnosed at stage I.

Sybil reported strong discrimination when examining baseline LDCTs across all participants, with AUCs ranging from 0.91 at 1 year (95% CI: 87.7–94.6) to 0.82 at 6 years (95% CI: 77.6–86.1) (Fig. 1*A*). For Black participants, AUCs ranged from 0.91 at 1 year (95% CI: 86.0–96.3) to 0.81 at 6 years (95% CI: 74.5–87.1) (Fig. 1*C*), and when follow-up LDCTs were included in addition to baseline LDCTs, AUC values ranged from 0.91 (95% CI: 87.2–95.0) to 0.81 (95% CI: 75.7–85.4) for years 1 and 6, respectively (Fig. 1*D*). For non-Black participants, AUCs ranged from 0.91 at year 1 (95% CI: 84.6–98.3) to 0.84 at year 6 (95% CI: 76.3–92.3) at baseline (Fig. 1*E* and Supplementary Table 1). AUCs for all scans are similar to baseline only scans (Fig. 1*B*, *D*, and *F*).

We further compared the discriminatory performance of Sybil between the Black and non-Black populations using DeLong's test for independent AUCs. For both the baseline and all-scans analyses, no statistically significant differences in AUC were observed between the two groups across any prediction horizon ($p > 0.05$ for all; Supplementary Tables 1 and 2). These findings indicate comparable model discrimination performance across racial subgroups.

Discussion

In this retrospective evaluation of a predominantly Black LCS cohort within a US safety-net health system that primarily serves populations experiencing socioeconomic disadvantage and barriers to healthcare access, the Sybil deep learning model reported strong, consistent performance in predicting future lung cancer risk across multiple time horizons. These findings extend prior validations of Sybil to a racially diverse US screening population and support its generalizability beyond predominantly White and Asian cohorts.

These findings address a critical gap in artificial intelligence (AI) validation by reporting consistent performance in a predominantly Black population—a population historically underrepresented in model development. This is particularly important given concerns about bias in healthcare AI models and the limited validation studies in populations historically affected by disparities in screening access, eligibility, and outcomes.[12] Accurate risk prediction in underrepresented populations is essential for equitable clinical translation. Sybil's performance suggests that AI-based risk stratification may complement existing screening strategies by enabling individualized, imaging-based risk assessment independent of traditional clinical criteria.

Although this study was conducted within the United States, Sybil may also have potential applications in international LCS programs. Because the model estimates future lung cancer risk from a single LDCT without requiring demographic information, smoking history, or country-specific screening eligibility criteria, it could provide a standardized imaging-based approach to risk

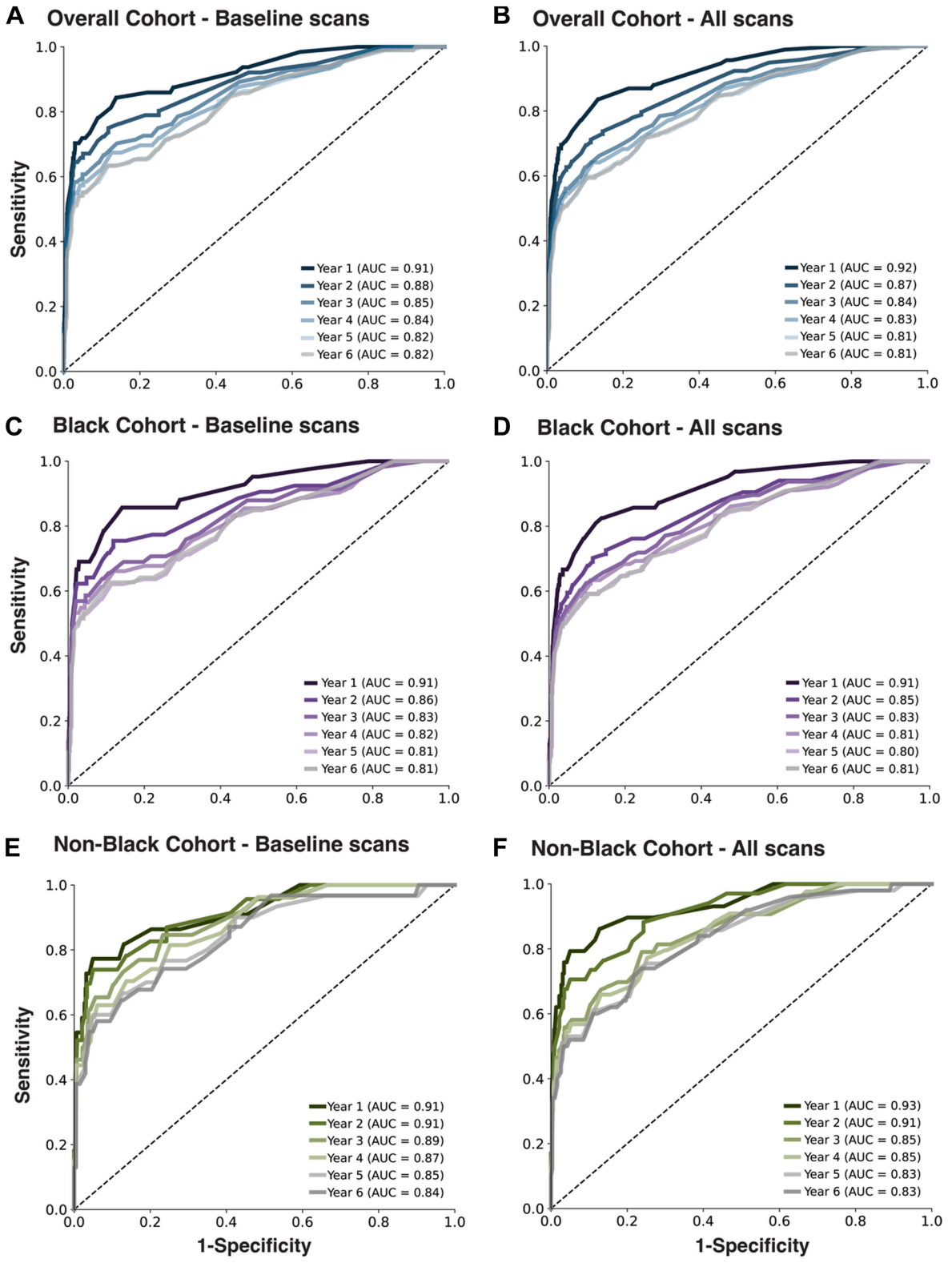


Figure 1. Sybil receiver operating characteristic curves for 1- to 6-year lung cancer risk prediction in the overall, Black, and non-Black cohorts. *(A)* Baseline-only LDCTs for the entire cohort (N = 3177). *(B)* All LDCTs for the entire cohort (N = 5883). *(C)* Baseline-only LDCTs for Black participants (n = 2061). *(D)* All LDCTs for Black participants (n = 3833). *(E)* Baseline-only LDCTs for non-Black participants (n = 1116). *(F)* All LDCTs for non-Black participants (n = 2050). AUC, area under the receiver operating characteristic curve; LDCT, low-dose computed tomography.

assessment across different healthcare settings. This may be particularly relevant in regions where smoking-based eligibility criteria do not adequately capture individuals at risk because of differences in tobacco exposure, sensitivity to tobacco carcinogens, environmental or occupational exposures, air pollution, or lung cancer risk among individuals who have never smoked.

Sybil-derived risk estimates could potentially inform screening eligibility, the interval between screening examinations, and the evaluation of individuals undergoing LDCT outside established screening programs. Nevertheless, differences in the computed tomography acquisition protocols among countries, screening infrastructure, and healthcare delivery may affect model performance and implementation. External validation in diverse international populations and healthcare systems is necessary and should be performed in a manner reflecting the intended clinical use before adoption.

This study was conducted in collaboration with the Sybil Implementation Consortium, a multi-institutional partnership including Massachusetts Institute of Technology, Mass General Brigham, the University of Illinois Chicago, Baptist Memorial Healthcare, and Wellstar Health System.[13] Ongoing studies within the Consortium are evaluating the integration of Sybil risk estimates into lung cancer early detection workflows and aim to assess how Sybil-derived risk estimates can inform clinical decision-making, screening eligibility and intervals, patient communication, and screening adherence.

Limitations include the retrospective, single-center design and reliance on self-reported smoking history, which may be subject to recall or reporting bias. In addition, the relatively high proportion of cancers diagnosed within 1 year may reflect the inclusion of prevalent cases, which could inflate short-term discrimination estimates. Furthermore, a subset of individuals in this cohort underwent screening outside traditional USPSTF criteria, reflecting real-world clinical practice.[14] The relatively small number of such patients precluded comparative analysis of Sybil's performance in this subset. Nonetheless, these findings support Sybil's performance in diverse screening populations.

CRediT Authorship Contribution Statement

Mary M. Pasquinelli: Conceptualization, Methodology, Investigation, Resources, Supervision, Project administration, Funding acquisition, Writing - original draft, Writing - review & editing.

Clinton H. Durney: Methodology, Software, Validation, Formal analysis, Data curation, Visualization.

Teri Hill: Investigation, Resources, Data curation, Project administration.

Ukamaka Atueyi: Investigation, Writing - review & editing.

Kevin Kovitz: Conceptualization, Resources, Supervision, Writing - review & editing.

Khaled Abdelhady: Investigation, Writing - review & editing.

Matthew Koshy: Investigation, Writing - review & editing.

Meida Wang: Validation, Formal analysis.

Raymond Osarogiagbon: Conceptualization, Methodology, Supervision, Writing - review & editing.

William R. Mayfield: Conceptualization, Methodology, Supervision, Writing - review & editing.

Erica T. Warner: Methodology, Formal analysis, Writing - review & editing.

Matthew P. Smeltzer: Methodology, Formal analysis, Validation, Visualization.

Osa Ogbeide: Investigation, Resources, Writing - review & editing.

Jill Feldman: Conceptualization, Writing - review & editing.

Alexandra L. Potter: Investigation, Data curation, Writing - review & editing.

Chi-Fu Jeffrey Yang: Methodology, Formal analysis, Supervision, Writing - review & editing.

Denise Kent: Investigation, Writing - review & editing.

Allison Chang: Methodology, Formal analysis, Writing - review & editing.

Florian J. Fintelmann: Conceptualization, Methodology, Validation, Supervision, Writing - review & editing.

Lecia V. Sequist: Conceptualization, Methodology, Supervision, Writing - review & editing.

Frank Weinberg: Conceptualization, Investigation, Methodology, Resources, Supervision, Funding acquisition, Writing - review & editing.

Disclosure

Dr. Pasquinelli reports receiving institutional funding from AstraZeneca Pharmaceuticals and Eli Lilly for the current work; travel support from the International Association for the Study of Lung Cancer to attend the World Conference on Lung Cancer; serving on the International Association for the Study of Lung Cancer Board of Directors; and holding a leadership role with the National Lung Cancer Roundtable. Dr. Durney reports receiving consulting fees from the University of Illinois Chicago Department of Medicine for computational workflow consulting. Dr. Atueyi reports receiving institutional funding from AstraZeneca and Eli Lilly for the current work. Dr. Osarogiagbon reports receiving grant funding from AstraZeneca; consulting fees from

AstraZeneca and Median Technologies; honoraria from Ideology Health; travel support from the Bristol Myers Squibb Foundation; holding a patent related to a lymph node specimen collection kit and method for lymph node examination; leadership roles with the American Cancer Society National Lung Cancer Roundtable, SWOG Research Network, and the Bristol Myers Squibb Foundation General External Advisory Council; and stock ownership in BridgeBio Pharma, BridgeBio Oncology Therapeutics, Erasca Inc., Gilead Sciences, Immunocore Holdings, Iovance Biotherapeutics, Eli Lilly, Pfizer, PMV Pharmaceuticals, Revolution Medicines, Tango Therapeutics, and Viatris. Alex Potter reports serving on the board of the American Lung Cancer Screening Initiative. Dr. Yang reports receiving a research grant from the Bristol Myers Squibb Foundation; consulting and advisory board fees from AstraZeneca and Genentech; honoraria from AstraZeneca; and serving on the board of the American Lung Cancer Screening Initiative. Dr. Fintelmann reports receiving grants or contracts from AstraZeneca, the Bristol Myers Squibb Foundation, and the National Institutes of Health/National Cancer Institute; consulting fees from Boston Scientific Corporation, Becton Dickinson, Single Pass Inc., Tubulis Inc., NeuWave Medical, and Johnson & Johnson MedTech; honoraria from Boston Scientific Corporation and Varian Medical Systems; travel support from Boston Scientific Corporation and Varian Medical Systems; participation on advisory or monitoring boards for WCG and Pfizer; equipment, materials, drugs, or services from Pfizer; an unrelated patent (US11322259B2/WO2019051358A1); and a leadership role with the Society of Interventional Oncology. Dr. Sequist reports receiving an institutional grant from AstraZeneca for the current work, an institutional grant from Delfi Diagnostics outside the current work, and serving as a board member of the International Association for the Study of Lung Cancer. Dr. Weinberg reports receiving support from AstraZeneca and Eli Lilly for the current work. The remaining authors declare no conflict of interest.

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Supplementary Data

Note: To access the supplementary material accompanying this article, visit the online version of the *Journal of Thoracic Oncology* at www.jto.org and at https://doi.org/10.1016/j.jtho.2026.106128.

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